



SUBJECT	PHYSICS – EXAMINATION
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mrs Govender
DATE	15 July 2025
MARKS	135
DURATION	120 minutes

%

LEARNER NAME:

TEACHER NAME:

MCCOY

HILDER

NTOMBELA

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 17 pages and a Data Sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your name and surname in the space provided above AND circle your teachers name.
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. Blank pages and blank graphs are included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

Question	1	2	3	4	5	6	7	8	Total
Mark									
Moderated Mark									
Question Total	22	17	21	18	19	20	9	9	135

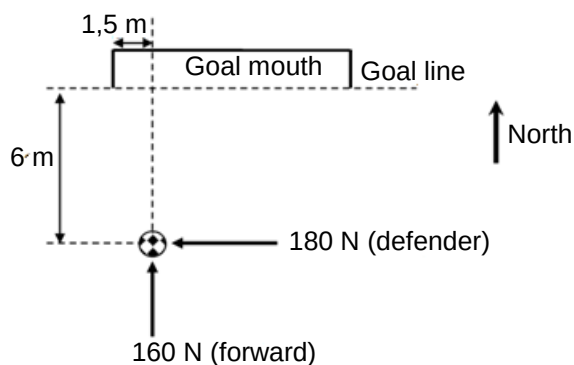
Comment:

Learner's Signature:

Teacher's Signature:

QUESTION 1**22 MARKS**

- 1.1 Two players kick a stationary soccer ball at the same time. The defender kicks with a force of 180 N due west while a forward kicks the ball with a force of 160 N due north as shown in the diagram. Assume that the time of contact between the ball and the players are the same.



- 1.1.1 Calculate the resultant force acting on the soccer ball while being kicked.
(Ignore the weight of the ball.) (4)

The ball is at the position shown in the diagram. The ball leaves the players at a speed of $18 \text{ m}\cdot\text{s}^{-1}$. The ball experiences frictional forces from the grass and the air and accelerates at $2,5 \text{ m s}^{-2}$.

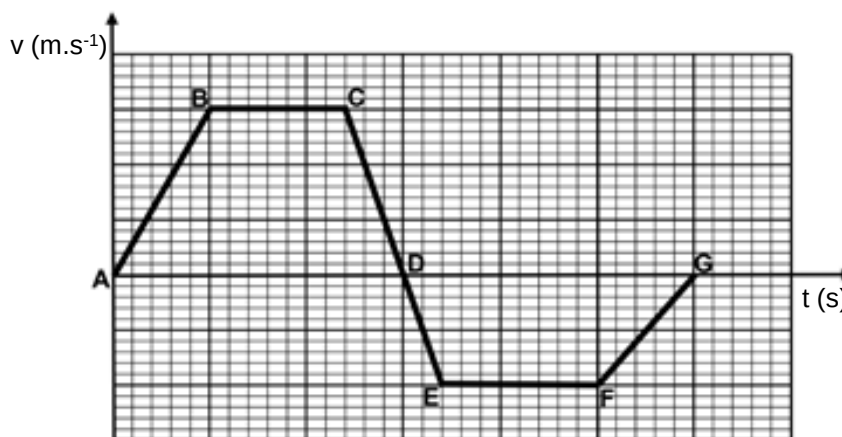
- 1.1.2 Calculate the distance travelled by the ball before it stops. (3)

In order to score a goal, the ball must go into the goal mouth and cross the goal line.

1.1.3 Will a goal be scored? Use a calculation to motivate your answer. (5)

1.2 A player runs in a straight line towards goal post to see if the ball crossed the goal line. After running some distance, the player turns and runs back (in a straight line) towards the starting point.

The graph below shows the velocity of the child against time.



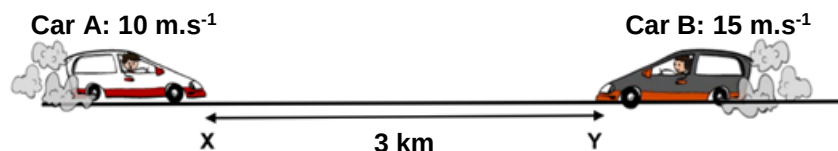
1.2.1 During which interval was the magnitude of the player's velocity the greatest? (1)

1.2.2 During which interval(s) was the child running south? (1)

1.2.3 During which interval was the magnitude of the child's acceleration greatest? (1)

1.3 Car A and Car B are travelling on a straight, level track. Points **X** and **Y** are 3 km apart on the track.

- Car A passes point **X** at precisely 10:00 travelling to the right at $10 \text{ m}\cdot\text{s}^{-1}$.
- Car B passes point **Y** at 10:02 (2 minutes after Car A passed point **X**), travelling to the left at $15 \text{ m}\cdot\text{s}^{-1}$.



Calculate the time on the clock when and the position where Car A and Car B will pass each other. Round the time off to the nearest minute and state the position in metres to the right of **X**. (7)

QUESTION 2

17 MARKS

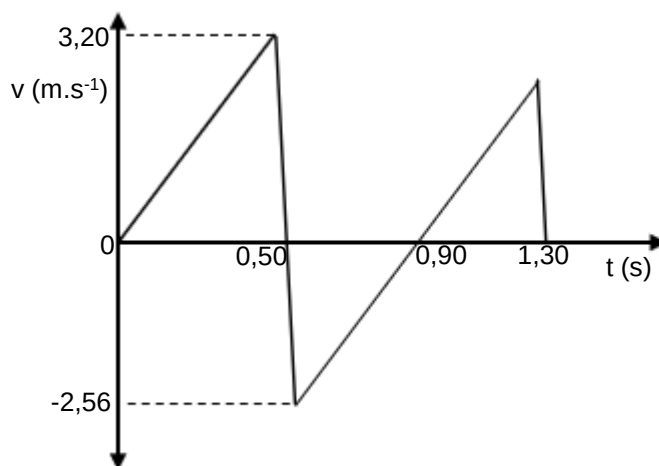
A St Benedict's matric boy arrived at the matric dance in a private helicopter. When the pilot was leaving the ground vertically upwards at a constant velocity. A watch is accidentally dropped from the helicopter when the helicopter is at a height of 52,2 m as shown in the diagram. The watch strikes the ground 6 s after it is dropped. Ignore air resistance.



- 2.1 Define the term *velocity*. (2)
- 2.2 Calculate the speed with which the helicopter is rising when the watch is dropped. (4)
- 2.3 State the velocity of the watch at the instant it is dropped. (2)
- 2.4 Calculate the maximum height that the watch reaches above the ground. (5)
- 2.5 Calculate the height of the helicopter when the cell phone strikes the ground. (4)

QUESTION 3**21 MARKS**

A ball with a mass of 2 kg on a distant planet is dropped from a certain height. The velocity-time graph represents the motion of the ball as it bounces vertically on a solid floor. The planet has no atmosphere so there is zero air resistance.



- 3.1 Define the term *acceleration*. (2)
- 3.2 Calculate the magnitude of the acceleration of the ball on this planet. (3)
- 3.3 Calculate the height the ball bounced to. (3)
- 3.4 Draw a labelled free-body diagram showing the forces acting on the ball at 0,7 s. Ignore air resistance. (2)

3.5 State the magnitude of the velocity with which the ball hits the ground at 1,30 s. (2)

3.6 Explain your answer in 3.5. (2)

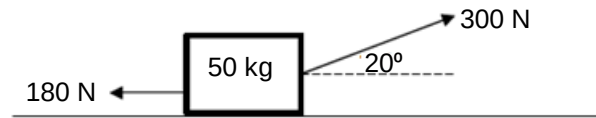
3.7 Determine the magnitude of a normal force experienced by the ball during the bounce, if the ball was in contact with the ground for 0,02 s. (4)

3.8 Draw a position-time sketch graph for the motion of the ball as it bounces vertically on a solid floor. Use the release point as the reference. (3)



QUESTION 4**18 MARKS**

A 50 kg metal box is pulled across a rough horizontal surface with a force of 300 N acting at an angle of 20° to the surface. The frictional force acting on the metal box is 180 N.



4.1 State *Newton's second law of motion*. (2)

4.2 Draw a free-body diagram showing all the forces acting on the metal box. (4)

4.3 Calculate the magnitude of the acceleration of the metal box. (5)

4.4 Calculate the magnitude of the normal force that the ground is exerting on the box. (4)

- 4.5 The box continues being pulled with a force of 300 N at an angle of 20° to the surface but the box now slides up an inclined plane.

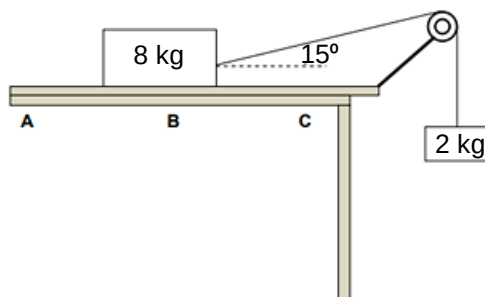
Will the frictional force acting on the metal box increase, decrease or stay the same?

Briefly explain your answer making use of a relevant equation.

(3)

QUESTION 5**19 MARKS**

A block, of mass 8 kg, is placed on a rough horizontal surface. The 8 kg block, which is connected to a 2 kg block by means of a light inextensible string passing over a light frictionless pulley, starts sliding from point A, as shown below.



- 5.1 The string exerts a horizontal force on the 8 kg block. According to Newton's Third Law, what is the reaction force to this? (2)
- 5.2 When the 8 kg block reaches point B, the angle between the string and the horizontal is 15° and the acceleration of the system is $1,32 \text{ m}\cdot\text{s}^{-2}$.
- 5.2.1 Give a reason why the system is NOT in equilibrium. (1)
- 5.2.2 Use the 2 kg mass to calculate the tension in the string. (3)
- 5.2.3 Calculate the kinetic frictional force between the 8 kg block and the horizontal surface. (4)

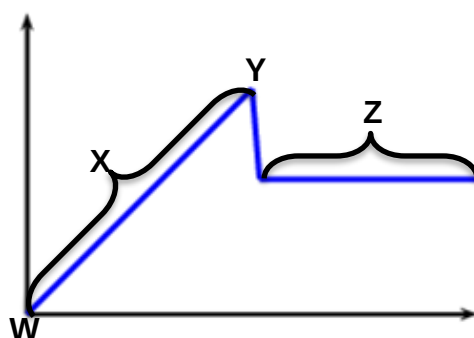
5.3 A coin was placed on the 8 kg block, which stays in place as the block was slides from B to C. Explain how it is possible for the coin to stay in place. (2)

5.4 The horizontal surface on which the 8 kg block is moving, is replaced by another horizontal surface made from a different material.

5.4.1 Will the kinetic frictional force, calculated in QUESTION 5.2.3 above, change?
Choose from: YES or NO. (1)

5.4.2 Give a reason for the answer. (2)

5.5 The graph below shows the motion of the block.



5.5.1 From which region is the block stationary? _____ (1)

5.5.2 From which point does the block start moving? _____ (1)

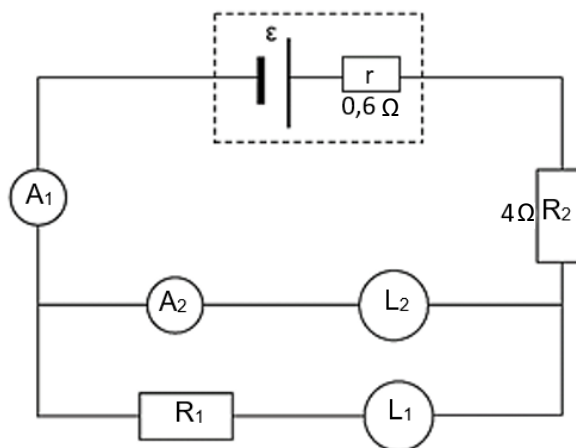
5.5.3 Explain your answer in 5.5.2 by referring to the types of friction acting. (2)

QUESTION 6**20 MARKS**

L_1 and L_2 are two light bulbs that have the following ratings:

L_1 : 36 W; 20 V and L_2 : 48 W; 32 V

The two bulbs are connected as shown in the circuit diagram below. The battery has an internal resistance of $0,6 \, \Omega$ while the conducting wires and the ammeters have negligible resistance. R_1 and R_2 are resistors.



6.1 Define the term *power*. (2)

6.2 Define the term *resistance*. (2)

6.3 If both light bulbs operate as RATED, calculate the:

6.3.1 Reading on ammeter A_2 . (3)

6.3.2 Reading on ammeter A_1 . (2)

6.3.3 Resistance of resistor R_1 . (3)

6.3.4 Emf of the battery. (4)

6.4 Bulb L_1 burns out after a while.

State what would happen to the power dissipated by R_2 . Use a suitable equation to help you explain your answer. (4)

QUESTION 7**9 MARKS**

7.1 An electrician needs to replace a bulb that has a 220 V supply. Unfortunately, the only working light bulb in stock is rated **50 V, 100 W**.

7.1.1 Calculate the resistance of the 50 V, 100 W light bulb. (3)

The electrician uses the 50 V, 100 W light bulb and connects a resistor in series with the light bulb to ensure that the bulb operates at its correct rating when connected to the 220 V supply.

7.1.2 What should the resistance of the resistor be that the electrician connects in series with the light bulb? (3)

7.1.3 Calculate the cost of using the light bulb for 45 minutes in a town where the price of electricity is R 1, 70 per kWh. (3)

QUESTION 8**9 MARKS**

- 8.1 The table below gives the frequency range of photons of different types of electromagnetic radiation.

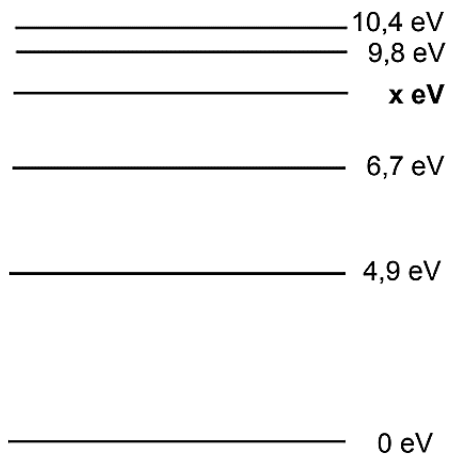
Electromagnetic radiation type	Frequency range of photons
Ultraviolet	Greater than $7,9 \times 10^{14}$
Violet – blue visible light	$6,11 - 7,9 \times 10^{14}$
Yellow – green visible light	$5,11 - 6,11 \times 10^{14}$
Red – orange visible light	$4,0 - 5,11 \times 10^{14}$
Infrared	Less than $4,0 \times 10^{14}$

- 8.1.1 Convert 2,2 eV to Joules. (2)

- 8.1.2 Calculate the frequency of a photon of radiation with energy 2,2 eV. (3)

- 8.1.3 What colour of light does a photon of 2,2 eV produce? (1)

- 8.2 In a sample of excited mercury atoms, all of the energy levels shown in the figure below are occupied. One of the energy levels in the figure is labelled **x eV**. The emission spectrum of mercury shows lines at approximately 0,9 eV, 1,5 eV and 2,2 eV.



Calculate the value of **x** in eV, showing all your working out.

(3)

ADDITIONAL SPACE

[illegible]